

AIDAR — PRODUCT REQUIREMENTS DOCUMENT

Product Name: Aidar **Domain:** getaidar.com **Legal Entity:** EML Global Ventures Limited (incorporation of Aidar Technologies Limited pending) **Document Status:** Version 3 — Living Document **Last Updated:** June 2026

1. EXECUTIVE SUMMARY

Aidar is a healthcare discovery and booking platform built for Lagos, Nigeria, with a roadmap for expansion across Nigeria and West Africa. The platform connects patients with verified medical practitioners and medical facilities for clinic visits, home visits, and virtual consultations.

Aidar's core differentiation is a trust and safety layer that does not exist in comparable Nigerian healthtech products: credential-verified practitioners, ID-verified patients for home visits, facility discovery with live maps and directions, end-to-end booking and payment, structured dispute resolution, booking-scoped in-app messaging, and transparent earnings tracking for practitioners.

The platform is migrating from a Lovable.dev/Supabase MVP to a production-grade self-hosted stack (Next.js 15 + NestJS + PostgreSQL/PostGIS) to support geo search, compliance requirements, and scale.

2. PROBLEM STATEMENT

Nigeria has approximately one doctor per 5,000 people against a WHO recommendation of one per 600. However, the deeper problem is not purely a shortage — it is a discovery and trust gap. Qualified, available practitioners exist across Lagos but patients have no reliable way to find, verify, or book them. Meanwhile practitioners with unfilled slots have no professional digital presence.

Existing Solution	Limitation
Helium Health	B2B hospital software, not consumer-facing
Kangpe / Wellvis	Teleconsultation only, no in-person discovery

MedxVerse	AI-driven teleconsultation, no practitioner/facility discovery, no home visits, no credential verification
Google Maps / word of mouth	No verification, no booking, no trust layer

3. USER TYPES

User Type	Description	Can Book?	Can Be Booked?
Patient	Searches for and books healthcare services	Yes	No
Practitioner	Independent medical professional listing services	No	Yes
Facility	Hospital, clinic, lab, pharmacy, or other medical facility	No	No (discoverable, contactable)
Admin	Platform operator managing verification, disputes, oversight	N/A	N/A

Practitioners and facilities retain full search and discovery access but cannot initiate bookings. This restriction is enforced on both frontend (hidden buttons) and backend (NestJS guards blocking non-patient JWT roles) so it cannot be bypassed.

4. AUTHENTICATION & ACCOUNT MANAGEMENT

4.1 Signup Flows

Authentication is handled via **JWT (access + refresh token)** with **argon2/bcrypt** password hashing via NestJS. No plain-text passwords are stored anywhere.

Patient Signup Fields: full name, email, phone number, password (all required) Flow: submit → email verification → login → patient dashboard

Practitioner Signup Fields: full name, email, phone number, specialty (dropdown), password (all required) Flow: email verification → login → 5-step onboarding

Facility Signup Fields: full name, email, phone number, password (all required) Flow: email

verification → login → 5-step onboarding

4.2 Login

- Patients → patient dashboard
- Practitioners → practitioner dashboard, or onboarding if incomplete
- Facilities → facility dashboard, or onboarding if incomplete
- Admins → `/admin` (NestJS guard; non-admins get 403)

4.3 Password Reset

- Forgot Password link → email from Resend via `notifications.getaidar.com`
- Token-based reset link (expiry: 1 hour)
- User sets new password → redirected to login

4.4 Field Validation

All fields are required with the following explicit exceptions:

- Practitioner: clinic/practice name; home visit fee (only if home visits enabled); virtual consultation fee and platform (only if virtual enabled)
- Facility: year established, website URL, HEFAMAA license number, CAC registration number

Required fields show a red asterisk. Submitting without them shows an inline “This field is required” and auto-scrolls to the first unfilled field.

Email: must contain `@`, valid domain, no spaces, min 2 chars before `@` and after final `.` — validated in real time with green tick on valid input.

Phone (Nigerian format): accepted formats `08XXXXXXXXXX`, `+2348XXXXXXXXXX`, `2348XXXXXXXXXX`; must begin with 070, 080, 081, 090, or 091; validated in real time.

4.5 Duplicate Prevention

The same email or phone number cannot be registered across any user type. Checked real time on field blur, case-insensitively for email, normalised across `0 / +234 / 234` for phone.

5. PRACTITIONER ONBOARDING (5-STEP FLOW)

Step 1 — Personal Information

Full name (pre-filled), phone number (pre-filled), WhatsApp number, profile photo upload (→ Cloudinary, public bucket), short bio (max 300 characters with live character count).

Step 2 — Professional Details

Specialty (dropdown — see Section 11), qualifications (e.g. MBBS, FMCP), regulatory body registration number, years of experience, credential certificate upload (→ Cloudflare R2 / S3 private bucket, signed URL access).

Note on verification language: The platform uses “credential verified against their relevant professional regulatory body” rather than “MDCN verified” because not all medical practitioners in Nigeria fall under MDCN. The verification system supports any applicable regulatory body.

Step 3 — Practice Details

Clinic/practice name (optional), Lagos area (dropdown — 20 areas), full address, consultation fee (Naira), home visits toggle → conditional home visit fee, virtual consultations toggle → conditional virtual fee + preferred video platform (Google Meet, Zoom, WhatsApp Video Call, Microsoft Teams, Any Platform).

Step 4 — Payment Details

Bank name (dropdown of 30 major Nigerian banks), account number (10 digits), account name (auto-populated via Paystack account resolution API, status shown). Privacy notice displayed. Bank details stored encrypted, visible only to authorised admin — never to patients or on the public profile.

Step 5 — Confirmation

Full summary, bank number masked to last 4 digits. On submit: `is_verified = false`, `status = pending`. Profile does not appear in public search until admin verifies.

6. FACILITY ONBOARDING (5-STEP FLOW)

Step 1 — Facility Information

Facility name, facility type (dropdown), description (max 500 characters), year established (optional).

Step 2 — Location Details

Full address, Lagos area dropdown, Google Maps pin (search → drop pin → lat/long saved via PostGIS `GEOGRAPHY(Point)` column), phone, WhatsApp, email, website URL (optional).

Step 3 — Facility Details

Opening hours, services offered (multi-select custom tags), emergency services toggle, ambulance toggle, NHIS acceptance toggle, **HMO/Insurance acceptance toggle** (see Section 29.1).

Step 4 — Documents and Photos

Facility logo (→ Cloudinary, public), up to 5 facility photos (→ Cloudinary, public), HEFAMAA license number (optional), CAC registration number (optional).

Step 5 — Confirmation

Summary of all details. On submit: `is_verified = false`, `status = pending`. Facility does not appear in public search until admin verification.

7. HOMEPAGE

7.1 Navigation Bar

Logo (links to homepage), links (Find a Doctor, Find a Facility, For Practitioners, For Facilities, About), Login / Sign Up. Mobile: hamburger menu.

7.2 Hero Section

Headline + subheadline, search bar with Practitioners ↔ Facilities toggle, specialty or facility type dropdown (conditional), Lagos area dropdown, Search button. Trust indicators below search bar.

7.3 How It Works

Three-step: search → view verified profiles → book or get directions.

7.4 Browse by Specialty

Clickable specialty cards. Each links to search filtered by that exact specialty value (database value = search filter value — no mismatch). See Section 11.

7.5 Browse by Facility Type

Clickable facility type cards → filtered facility search.

7.6 For Practitioners / For Facilities

Benefits + respective CTA → signup flows.

7.7 Footer

Logo, About, Contact, Privacy Policy, Terms of Service, Safety Centre, For Practitioners, For Facilities. Copyright: "© 2026 Aidar — EML Global Ventures Limited. All rights reserved."

8. SEARCH, DISCOVERY & GEO INFRASTRUCTURE

8.1 Geo Infrastructure (PostGIS)

Practitioner and facility locations are stored as `GEOGRAPHY(Point, 4326)` columns in PostgreSQL via PostGIS. This enables:

- **"Near Me"** — true radius-based geo queries (`ST_DWithin`) ordered by distance from the user's GPS coordinates
- **Area filters** — fast spatial filtering by Lagos area polygon or bounding box
- **Specialty + proximity combined** — e.g. "Cardiologists within 5km of my location"
- **Distance sorting** — results ordered by actual geodesic distance, not just text-matched area

At current scale, PostGIS handles all search needs. **Meilisearch or Typesense** will be introduced post-launch when listing volume makes full-text search necessary — designed in as a deliberate future integration, not a retrofit.

8.2 Practitioner Search Results

Each card shows: profile photo, full name, specialty, Lagos area, consultation fee, years of experience, verified badge, home visits badge (if applicable), virtual badge (if applicable), star rating + review count, response time, acceptance rate (see Section 29.4), Book Now button.

Filters: Verified only, Offers home visits, Offers virtual consultations, Consultation fee range, **Accepts HMO/Insurance** (see Section 29.1) **Sort:** Rating, Fee (low–high), Fee (high–low), Distance (Near Me), Newest

Only verified practitioners (`is_verified = true`) appear in results.

8.3 Facility Search Results

Each card shows: facility logo, name, facility type badge, Lagos area, opening hours, first 3 services (+X more), verified badge, emergency services badge, NHIS badge, **HMO/Insurance badge** (if applicable), Call/WhatsApp/View Profile buttons.

Filters: Facility type, Has emergency services, Accepts NHIS, **Accepts HMO/Insurance**, Verified only

Only verified facilities appear in results.

8.4 Map View

- Toggle between List View and Map View on both tabs
- Google Maps with PostGIS-backed pins for all results
- Tapping a pin shows: name, type, geodesic distance, ETA by car and on foot, View Profile, Directions (opens Google Maps app)
- “Near Me” button recenters map and re-runs geo query from user’s location
- Graceful fallback if location access denied

9. PRACTITIONER PUBLIC PROFILE

Section	Contents
Hero	Photo, name, specialty, verified badge, star rating + review count, response time, acceptance rate (Section 29.4), Lagos area, years of experience, regulatory body registration number
About	Bio, qualifications
Consultation Details	Clinic visit fee; home visit fee (if offered); virtual consultation fee + platform (if offered)
Booking Buttons	Book Clinic Visit (always); Book Home Visit (only if <code>offers_home_visits = true</code>); Book Virtual Consultation (only if <code>offers_virtual = true</code>) — all hidden for logged-in practitioners/facilities
Location	Lagos area, address, embedded Google Map (general area), Get Directions
Nearby Practitioners	Up to 3 other verified practitioners in same area + specialty; hidden if none

Reviews	Average rating, total count, star-by-star breakdown, sort (Most recent / Highest / Lowest), individual reviews show patient first name only, rating, text, time posted, booking type
Share	Copy-to-clipboard share button

Practitioner profile photo is clickable in the chat header on the patient side (see Section 17.5).

10. FACILITY PUBLIC PROFILE

Section	Contents
Hero	Logo, cover photo, name, facility type badge, verified badge, emergency services badge, NHIS badge, HMO/Insurance badge (if applicable), Lagos area, address, opening hours
Photo Gallery	Grid of up to 5 uploaded photos, tap to enlarge on mobile
About	Description, year established, services offered as tags
Location & Directions	Embedded Google Map with PostGIS-precise pin; Drive tab (ETA, distance, route); Walk tab (ETA, distance); Open in Google Maps; Refresh My Location; fallback if location denied
Contact	Call, WhatsApp (pre-filled message), Email, Website (if provided)
Nearby Facilities	Up to 3 verified facilities in same area

11. SPECIALTY LIST

Practitioner-title naming (not field names) used consistently across signup, onboarding, search, homepage cards, and database. The value stored in the database matches exactly the value used in search filtering — no mismatch.

Doctor, Dentist, Nurse, Midwife, Pharmacist, Physiotherapist, Optometrist, Psychiatrist, Psychologist, Counsellor, Cardiologist, Dermatologist, Paediatrician, Gynaecologist, Obstetrician, Surgeon, Orthopaedic Surgeon, ENT Specialist, Neurologist, Urologist,

Gastroenterologist, Pulmonologist, Endocrinologist, Oncologist, Nephrologist, Rheumatologist, Haematologist, Radiologist, Anaesthetist, Dietitian, Nutritionist, Speech Therapist, Occupational Therapist, Medical Laboratory Scientist, Radiographer, Community Health Officer, Mental Health Nurse, Fertility Specialist, Plastic Surgeon, Ophthalmologist

Homepage features: Doctor, Dentist, Physiotherapist, Pharmacist, Optometrist, Psychiatrist, Paediatrician, Gynaecologist, Cardiologist, Dermatologist, Nutritionist, Dietitian, Surgeon, Neurologist, Mental Health Nurse, Fertility Specialist.

12. BOOKING SYSTEM

12.1 Access Restriction

Only `user_type = "patient"` JWT-authenticated users can initiate bookings. Booking buttons are hidden for practitioners/facilities on the frontend. The NestJS booking controller enforces this via role guards — non-patient JWT tokens receive 403 regardless of frontend state.

12.2 Date and Time Slot Selection

30-minute slots grouped into three collapsible periods:

- **Morning (8:00am–11:30am):** 8:00, 8:30, 9:00, 9:30, 10:00, 10:30, 11:00, 11:30
- **Afternoon (12:00pm–3:30pm):** 12:00, 12:30, 1:00, 1:30, 2:00, 2:30, 3:00, 3:30
- **Evening (4:00pm–7:00pm):** 4:00, 4:30, 5:00, 5:30, 6:00, 6:30, 7:00

4-column grid on desktop, 3-column on mobile. Selected slot highlighted in primary teal.

Summary above Confirm: "Selected: [date] at [time]".

- Already booked slots: grey with strikethrough, unselectable
- Today: past slots hidden; if none remain, prompt to choose a different date
- Sundays: no slots shown, message displayed

12.3 Database-Level Booking Concurrency

Double-booking prevention is enforced at the **database level** — not just in application code — to eliminate race conditions when two patients attempt the same slot simultaneously.

A **unique partial index** on the `bookings` table:

```
CREATE UNIQUE INDEX bookings_practitioner_slot_unique
```

```
ON bookings (practitioner_id, booking_date, time_slot)
WHERE status NOT IN ('cancelled', 'cancelled_no_show');
```

This means the database itself rejects a second insert for the same practitioner + date + slot as a constraint violation, regardless of which application server or request processed the payment first. Application code catches this constraint error and returns the user to slot selection with: *“Sorry, this slot was just booked by another patient. Please select a different time slot.”* The conflicting slot is immediately shown as unavailable.

This replaces the previous application-level race condition described in earlier versions of the PRD.

12.4 Practitioner Availability & Calendar Scheduling

Accessible via the “Manage Availability” button on the practitioner dashboard.

Working Days: Seven toggles (Mon–Sun). Disabled days are greyed out and unselectable in the patient booking calendar. Saved to `availability_settings.working_days`.

Working Hours: Start time and End time dropdowns (30-minute increments, 6:00am–10:00pm). End must be after start. Saved to `availability_settings`.

Block Specific Dates: Single date + optional reason, or date range. Saved to `blocked_dates`. Displayed as grouped range entries with single “Unblock All” action. Blocked dates are greyed out in the patient booking calendar with tooltip: “Not available on this date.”

Non-breaking fallback: if a practitioner has not configured availability, all dates remain selectable — existing booking flow is unaffected.

12.5 Clinic Visit Booking

1. Patient clicks Book Clinic Visit (login required — NestJS guard)
2. Modal shows: practitioner photo/name/specialty, “Clinic Visit” label, fee, date picker (respects availability + blocked dates), 30-minute slot grid, optional notes (max 300 chars), pre-filled patient name/phone, “Payment Secured by Paystack” badge
3. Confirm Booking → Paystack popup → payment with `practitioner_id`, `patient_id`, booking date/time/type, and amount in Paystack metadata
4. Paystack webhook received by NestJS → booking created post-webhook (not on redirect) — eliminates cross-device failures caused by relying on redirect state
5. **On success:** booking saved (`status = pending`, `payment_status = paid`); confirmation screen with “**Message [Practitioner Name]**” button opening in-app chat

(Section 17); **SMS confirmation sent via Termii** (Section 23.2)

6. **On failure:** booking not saved; error: *"Payment was not completed. Please try again."*

12.6 Home Visit Booking

ID Verification Gate:

BVN removed. BVN is high-risk, tightly regulated by the CBN, and unnecessary when National ID, International Passport, or Driver's License already provides sufficient identity verification. Collecting BVN would require additional CBN compliance obligations disproportionate to the use case.

Accepted ID types: National ID (NIN), Driver's License, International Passport, Voter's Card.

Patient submits: ID type, ID number, ID photo upload (→ Cloudflare R2 private bucket, signed URL access), selfie with ID (→ Cloudflare R2 private bucket, signed URL). Full home address + nearest landmark (optional). Required consent checkbox (see NDPA Consent Framework, Document 3 of the Compliance Package).

On submit: status message shown. Admin reviews via ID verification queue. Patient receives **SMS notification** when ID is approved or rejected (Section 23.2).

Once verified: booking modal proceeds with home visit fee, home address field, landmark, safety notice, same slot/availability/payment flow as clinic visit.

File storage decision — ID documents and credential certificates are stored in a private encrypted bucket (Cloudflare R2 / S3) separate from the public media pipeline (Cloudinary). Access to private documents uses short-lived signed URLs generated server-side by NestJS. Profile photos, facility photos, and logos use Cloudinary (public CDN). This is a deliberate architectural decision, not a default.

12.7 Virtual Consultation Booking

Modal shows: Virtual Consultation label with camera icon, fee, video platform, date/time selection, notes field, pre-filled name/email. On success: booking saved (`type = virtual_consultation`); message: *"Virtual consultation booked! The practitioner will send you a meeting link at least 1 hour before your appointment."* Patient dashboard shows "Awaiting Meeting Link." Once practitioner adds link: "Join Consultation" button activates with countdown timer. **SMS and WhatsApp notification sent when meeting link is added** (Section 23.2).

12.8 Booking Lifecycle & Statuses

`pending → confirmed → in_progress → awaiting_confirmation → completed`

Alternate paths: `cancelled`, `cancelled_no_show`, `disputed`, `requires_review`

Auto-progression is handled by a **BullMQ scheduled job** (replaces `pg_cron`) running every 15 minutes on the NestJS backend, secured via internal queue authentication.

12.9 Cancellation Policy & Paystack Refund Flow

Timing	Patient Refund	Practitioner Compensation
>2 hours before appointment	Full refund of consultation fee	None
<2 hours before appointment	No refund	50% of consultation fee
Practitioner cancels	Full refund (consultation fee + booking fee)	None

Refund flow: On cancellation where a refund is due, the NestJS backend calls the Paystack Refunds API (`POST /refund`) with the original transaction reference and the refund amount. For full refunds, the full consultation fee is refunded. For partial refunds (dispute resolution), the specific amount is specified. The ₦1,000 booking fee is non-refundable in all patient-initiated cancellations. Paystack processes the refund to the patient's original payment method. Refund status is tracked in the `bookings` table (`refund_status` , `refund_reference`). Patients are notified of refund initiation via email and SMS.

12.10 Booking Escrow Position

Holding patient payment and releasing it to practitioners after a completed visit means Aidar is operationally functioning as a payment intermediary. At current volume this is handled manually, but as scale grows this carries CBN and Paystack Terms implications.

Immediate action (pre-scale): Ensure Paystack account type and agreement covers the marketplace/platform model being used. Review Paystack's SubAccounts feature (automated splits) to determine whether implementing it earlier than planned eliminates the escrow classification risk. This item must be resolved before or alongside Seed stage growth.

Target state: Automated Paystack SubAccount splits (85% practitioner / 15% Aidar) triggered on booking confirmation, not manually after completion. This removes the floating period and eliminates manual reconciliation.

12.11 Raise Dispute

Available from patient dashboard (hidden on already-disputed, completed, or cancelled bookings; one dispute per booking).

Modal: booking details + reason dropdown (Practitioner did not show up / Practitioner was late / Service not as described / I feel unsafe / Payment issue / Other) + optional details field (max 500 chars). On submit: `status = disputed`, payout frozen, patient sees: *"Your dispute has been received. Our team will contact you within 2 hours."* Admin, practitioner, and patient notified via email and SMS.

Admin resolution actions: Release to Practitioner, Refund Patient (calls Paystack Refunds API), Partial Resolution (calls Paystack Refunds API with specified partial amount).

13. PATIENT DASHBOARD

Element	Description
Welcome message	Patient's first name
Search bar	Toggle Practitioners / Facilities
Tabs	Upcoming Bookings (default), Booking History
Upcoming Bookings	Practitioner photo/name, specialty, date/time, type icon, status badge, amount paid; virtual: Join Consultation / Awaiting Meeting Link / countdown; Cancel button (if >2 hrs); Raise Dispute
Booking History	Practitioner photo/name, specialty, date/time, type, status, amount, reference, Leave Review button (for completed without review)
Review Prompt	Post-completion card: "How was your experience with [Name]? Leave a review"
Quick Actions	Find a Doctor, Find a Facility, Edit Profile
Chat Access	Floating icon bottom-left — see Section 17

Stats cards are practitioner-only and do not appear here.

14. PRACTITIONER DASHBOARD

Element	Description
Profile Overview	Photo, name, specialty, verification status badge
Stats Cards	Total Bookings, Pending, Completed, Cancelled
Tabs	Upcoming Bookings, Booking History
Upcoming Bookings	Patient name, date/time, type, amount, status; for home visits: Patient Verified badge + verified contact details; action buttons per status/type: Confirm, Start Visit, End Visit, Mark Complete, Cancel; virtual: Add Meeting Link; "Message Patient" button
Booking History	Patient name, date/time, type, status (completed/cancelled/disputed), amount, reference
Reviews Tab	Average rating summary + star breakdown; reviews list (patient first name, rating, text, time, booking type); sort options; unread badge
Availability Manager	Working days/hours + blocked dates — see Section 12.4
Earnings Tab	Lifetime earnings + date-range filter + itemised list — see Section 18.1
Payouts Tab	Read-only payout history — see Section 18.2
Chat Access	Floating icon bottom-left — see Section 17
Quick Actions	Edit Profile, View Public Profile

15. FACILITY DASHBOARD

Element	Description

Header	Facility logo, name, type, verification status badge
Stats Cards	Total profile views, total enquiries, total WhatsApp contacts
Quick Actions	Edit Facility Profile, View Public Profile, Upload New Photos
Tips Card	Onboarding tips for new facilities

Facilities do not have access to chat, earnings/payouts, or availability manager.

16. REVIEWS SYSTEM

- Triggered after booking is marked `completed`
 - Modal: 1–5 star selector (interactive), optional text (max 300 chars), Submit
 - One review per booking; patient first name only displayed
 - Immediately reflected on public profile average and search card rating
 - Sortable on public profile and practitioner Reviews tab (Most Recent / Highest / Lowest)
 - **Admin review moderation:** Admin can remove any review that is abusive, defamatory, fraudulent, or in violation of the Terms of Service. Removed reviews are soft-deleted (retained in `reviews` table with `status = removed` and `removed_by` admin ID for audit trail). The practitioner and patient are notified via email when a review is removed. This is a legal necessity — leaving abusive or defamatory content live without a moderation mechanism creates liability. See Section 19.6.
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17. MESSAGING & CHAT SYSTEM (PATIENT ↔ PRACTITIONER)

17.1 Conversation Model

One persistent conversation per patient–practitioner pair (enforced by `UNIQUE(patient_id, practitioner_id) on conversations`). Created on first booking; all subsequent bookings between the same pair reuse the existing conversation and its full history. Real-time delivery via **Socket.io** on the NestJS backend.

17.2 Floating Chat Icon

Bottom-left, all pages, visible to logged-in patients and practitioners only. Unread count badge. Opens slide-in panel (desktop) or full-screen (mobile).

17.3 Active vs Read-Only

An active booking = status `pending`, `confirmed`, `in_progress`, or `awaiting_confirmation` between this patient and practitioner.

- **At least one active booking exists:** input field enabled
- **No active booking exists:** input hidden, banner: *"This conversation is closed. You can still view your chat history. You'll be able to message again once you have a new booking with this practitioner."*

Checked on open and updated live via Socket.io if booking status changes mid-conversation.

17.4 Chat List View

Patient side: one row per practitioner ever booked — photo, name, specialty, last message preview, timestamp, unread indicator, Active/Closed label. Practitioner side: one row per patient ever booked — same structure. Sorted by most recent activity.

17.5 Chat Window

Header: other party's photo (clickable on patient side → practitioner public profile; not clickable on practitioner side), name, specialty (patient side), Active/Closed status. Full chronological history. Input supports up to 1,000 characters.

17.6 Entry Points

"Message [Practitioner Name]" — on booking confirmation screen after payment. Replaces WhatsApp redirect. Opens chat pre-filled with booking details (editable, not auto-sent).

"Message Patient" — on each upcoming booking card in practitioner dashboard.

WhatsApp contact buttons on facility profiles are unaffected and remain.

17.7 Real-Time & Read Status

Socket.io subscriptions: new messages appear instantly; unread counts update live; active/closed status updates live. Opening a conversation marks all other party's messages as read (`read_at` populated).

18. EARNINGS & PAYOUTS SYSTEM (PRACTITIONER)

DASHBOARD)

18.1 Earnings Tab

Lifetime earnings (sum of `completed` + `paid` bookings), date-range picker filter, itemised list for selected period (date, patient name, booking type, amount). Plain styling — no celebratory design.

18.2 Payouts Tab

Read-only table: Date, Amount, Reference. Empty state: *"No payouts have been processed yet. Payouts are processed manually by our team."*

18.3 Admin-Side Payout Recording

On practitioner detail page in admin: "Record Payout" form (amount, date, reference/note). Entries appear immediately in practitioner's Payouts tab. Used for manual bank transfer logging until Paystack SubAccount automation is implemented.

19. ADMIN DASHBOARD

Accessible only at `/admin`; NestJS admin guard returns 403 for all other roles.

19.1 Overview Tab

Stats cards: total/verified/pending/rejected practitioners, total/verified facilities, total/verified patients, total bookings, total revenue. Alerts: high cancellation rates, disputes, `requires_review`, pending IDs >24 hours.

19.2 Practitioners Tab

Pending verification queue: full details + credential document (signed URL), Verify / Reject / Request More Info. Approved table: detail page with booking stats, full history, performance metrics, bank details, Record Payout form, admin actions (Suspend, Remove, Send Warning, Mark as Featured), CSV export. Rejected table with Reinstate.

19.3 Facilities Tab

Mirrors practitioners tab structure.

19.4 Patients Tab

ID Verification Queue: name, email, phone, ID type/number, ID document (Cloudflare R2

signed URL, full-size viewable), selfie (signed URL), submission date. Actions: Approve / Reject (requires reason) / Flag as Suspicious.

All Patients table: detail page per patient with booking stats, history, risk assessment (cancellation rate, dispute rate, late cancellation rate, risk level), ID verification details, admin actions.

19.5 Bookings Tab

Full bookings table with filters (status, type, date range, area). Disputes section (red, full details, resolution actions calling Paystack Refunds API). Requires Review section.

19.6 Reviews Moderation Tab

All reviews on the platform, filterable by: all reviews, flagged reviews, removed reviews. Admin can view the full review text, the booking it was tied to, and the patient and practitioner involved. **Remove Review** action: soft-deletes the review (`status = removed` , `removed_by` admin ID, `removal_reason` field), removes it from the practitioner's public profile and search card rating, and sends email notifications to both parties. This is a legal necessity — publishing defamatory or abusive content without a moderation mechanism creates liability for Aidar.

19.7 Audit Log

All admin actions are recorded in an `admin_audit_log` table: `admin_id` , `action_type` , `target_entity` (e.g. practitioner, booking, review), `target_id` , `payload_snapshot` , `timestamp` . This log is immutable (append-only) and cannot be deleted by admin users. It is visible to admins in a read-only Audit Log tab, filterable by action type, admin, and date range.

19.8 Bank Details (Payout Reference)

Practitioner bank details surfaced in: verification queue, practitioner detail page, Bookings tab (for manual payout reference on completed bookings), CSV exports. Never visible to patients or in any patient-facing interface.

20. SAFETY CENTRE (/safety)

For Patients: Only credential-verified practitioners on the platform. Payment secured before visit. Rate and review after every visit. Cancel >2 hours before for full refund.

For Practitioners: All home visit patients are ID verified. Full verified patient details before

accepting. Can decline any booking without penalty. Payment guaranteed before home visit departure. SOS button during visits. Live location sharing with emergency contact.

Virtual Consultations: Payment secured until consultation complete. Meeting link sent ≥ 1 hour before session. Cancel >2 hours before for full refund.

21. STATIC PAGES

- **Privacy Policy** (`/privacy-policy`) — NDPA/NDPR compliant, covers all data categories handled by the platform
- **Terms of Service** (`/terms`) — covers all booking types, cancellation, home visit, virtual, chat, reviews, HMO/insurance

Both linked from footer on every page.

22. PAYMENTS & MONETIZATION

22.1 Payment Processing

Paystack integration (test mode currently; live keys on public launch). Booking metadata (`practitioner_id` , `patient_id` , `date` , `time` , `type` , `amount`) is embedded in Paystack transaction metadata and confirmed via **Paystack webhook** received by NestJS — bookings are created post-webhook, not on redirect, eliminating cross-device failures.

Paystack refund flow: Full and partial refunds issued via Paystack Refunds API (`POST /refund`). $\$1,000$ booking fee is non-refundable on patient-initiated cancellations. Refund status tracked in `bookings` table. Patient notified via email and SMS.

Paystack SubAccount splits (automated 85/15): Currently manual. Target: implement Paystack SubAccounts so the practitioner's 85% is routed directly to their verified bank account at booking confirmation, eliminating the manual payout process and the escrow-position risk (see Section 12.10). This is now a **high-priority roadmap item, not a post-Seed afterthought**.

22.2 Revenue Model

Stream	Mechanism	Stage
		From

Patient booking fee	₦1,000 per confirmed booking	launch
Platform commission	15% of consultation fee	From launch
Practitioner premium subscription	₦10,000/month (priority listing, analytics)	Year 2+

Unit economics: ₦8,000 consultation → ₦1,200 commission + ₦1,000 booking fee = ₦2,200 platform revenue per booking.

23. NOTIFICATIONS — EMAIL, SMS & WHATSAPP

23.1 Channel Strategy

Email alone is insufficient for a Lagos-focused platform. Email open rates in Nigeria are weak and deliverability is variable. **SMS and WhatsApp are the primary channels through which users in Lagos are reliably reached.** The notification strategy therefore treats SMS as core from launch, with WhatsApp to follow.

Channel	Provider	Priority	Timeline
Email	Resend (<code>notifications.getaidar.com</code>)	Secondary (verification, documents)	Live
SMS	Termii or Africa's Talking	Primary (booking events, time-sensitive)	Launch
WhatsApp	WhatsApp Business API	Primary (rich messaging, links)	Post-launch

23.2 Notification Event Matrix

Event	Email	SMS	WhatsApp
Signup confirmation	✓	✓	—
Password reset	✓	—	—
Practitioner application received	✓	✓	—

Practitioner approved / rejected	✓	✓	—
Practitioner suspended / removed	✓	✓	—
Facility application received/approved/rejected	✓	✓	—
Patient ID verification submitted	✓	✓	—
Patient ID approved / rejected	✓	✓	—
Booking confirmed (patient)	✓	✓	✓ (post-launch)
New booking received (practitioner)	✓	✓	✓ (post-launch)
Practitioner confirmed booking	✓	✓	—
Visit started / "Practitioner on the way"	—	✓	✓ (post-launch)
Visit ended / awaiting confirmation	✓	✓	—
Booking completed + review prompt (patient)	✓	✓	—
Booking cancelled (both parties)	✓	✓	—
Refund initiated	✓	✓	—
Meeting link added (virtual)	✓	✓	✓ (post-launch)
Dispute raised (patient, practitioner, admin)	✓	✓	—
Review received (practitioner)	✓	—	—
Review removed (both parties)	✓	—	—
New unread message	In-app badge	—	—
Admin: new practitioner/facility/ID/dispute	✓	—	—
Referral reward earned	✓	✓	—

23.3 SMS Implementation

SMS delivered via **Termii** (Nigerian provider, supports OTP and transactional SMS) or **Africa’s Talking** (regional fallback). NestJS SMS service abstracts the provider so switching requires no API changes. Messages are concise (<160 chars where possible), include the Aidar name, and include the booking reference.

23.4 WhatsApp Implementation (Post-Launch)

WhatsApp Business API via a registered provider. Requires WhatsApp Business Account approval. Messages use approved template messages for booking confirmations, meeting links, and “practitioner on the way” alerts. Implementation deferred to post-launch due to registration lead time, but architected into the notification service from day one.

24. DATA PROTECTION & REGULATORY COMPLIANCE

This section reflects the requirements of the **Nigeria Data Protection Act 2023 (NDPA)** and the **Nigeria Data Protection Regulation 2019 (NDPR)**, enforced by the **Nigeria Data Protection Commission (NDPC)**. Aidar handles multiple categories of sensitive personal data — government ID documents, health-related consultation data, biometric selfies, and financial data — placing it in the highest-risk tier for Nigerian data regulation. Compliance is not optional.

A full compliance package has been prepared covering all documents below. See the separate Aidar Compliance Package document.

24.1 Data Categories and Sensitivity Classification

Data Category	Users Affected	Classification	Lawful Basis
Name, email, phone	All	Standard personal data	Contractual necessity
Profile photos	Practitioners, facilities	Standard personal data	Contractual necessity
Government ID documents + selfies	Patients (home visit)	Sensitive — biometric	Explicit consent + Legitimate interest
Health-related notes, specialties, consultation history	Patients	Sensitive — health	Explicit consent
Regulatory body credentials +		Standard +	Contractual

certificates	Practitioners	professional	necessity
Bank account details	Practitioners	Sensitive — financial	Contractual necessity
Real-time GPS location	All (maps feature)	Standard personal data	Consent (browser permission)
In-app chat messages	Patients, practitioners	Standard personal data	Contractual necessity
Waitlist data	Waitlist users	Standard personal data	Consent

BVN is explicitly excluded. BVN collection would require CBN compliance obligations disproportionate to the use case. National ID, Driver's License, International Passport, and Voter's Card provide sufficient identity verification for home visit booking.

24.2 Consent Framework

Separate, dedicated, non-pre-ticked consent checkboxes are required at the following points:

At patient signup (both required, one optional):

- Terms and Privacy Policy acceptance (required)
- Explicit consent to health-related data processing (required)
- Marketing communications consent (optional)

At home visit ID verification screen (required before any document upload):

- Explicit consent to identity document and selfie collection and processing, including disclosure to admin, purpose, retention period, and right to withdraw

At practitioner signup:

- Terms and Privacy Policy acceptance (required)
- Public profile consent — consent to professional data being publicly displayed (required)
- Bank details processing consent — specific consent covering encryption, admin-only access, payout purpose (required at onboarding Step 4)

At waitlist registration:

- Launch notification consent (required)

24.3 Data Storage — Public vs Private Split

Deliberate architectural decision: Sensitive documents are not stored in the same pipeline as public media.

File Type	Storage	Access
Profile photos, facility photos, logos	Cloudinary (public CDN)	Public URL
Practitioner credential certificates	Cloudflare R2 / S3 (private bucket)	Short-lived signed URL via NestJS
Patient ID documents	Cloudflare R2 / S3 (private bucket)	Short-lived signed URL via NestJS, admin only
Patient selfies	Cloudflare R2 / S3 (private bucket)	Short-lived signed URL via NestJS, admin only

Signed URLs expire after a short period (e.g. 15–60 minutes) and are generated server-side on authenticated request. No direct public URL exists for any sensitive document.

24.4 Data Retention Schedule

Data Category	Retention Period
Active account data	Duration of account
Deleted account data	90 days post-deletion, then purge
ID documents and selfies	12 months from last home visit booking
Booking records	2 years
Chat messages	12 months post-booking closure
Practitioner bank account details	Duration of listing + 6 months
Credential documents	Duration of listing + 12 months
Waitlist data	Until launch notification + 90 days
Payment/financial records	7 years (Nigerian tax law)

24.5 Data Protection Officer (DPO)

A DPO must be designated before public launch. At early stage this is the founder, with a dedicated privacy email (privacy@getaidar.com). Post-Seed: outsource to a Nigerian data protection compliance firm (typically ₦50,000–₦150,000/month).

24.6 NDPC Registration

Required once Aidar processes data of more than 1,000 data subjects in any rolling 6-month period. Given waitlist targets, this threshold is likely to be reached before or shortly after launch. Registration must be filed before crossing the threshold. Budget: ₦200,000–₦500,000. Full registration checklist is in the Compliance Package.

24.7 Data Breach Response

The NDPA requires NDPC notification within **72 hours** of becoming aware of a breach likely to affect data subjects. A full Breach Response Plan is included in the Compliance Package. Every admin action affecting user data is captured in the audit log (Section 19.7).

24.8 Data Processor Agreements

Written Data Processing Agreements (DPAs) are required with: Supabase (or self-hosted PostgreSQL provider), Paystack, Google Maps Platform, Resend, Cloudinary, Cloudflare R2 / AWS S3, Termii/Africa's Talking, Sentry. All DPAs must be accepted and confirmation saved before public launch.

24.9 Compliance Documents (Separate Package)

The following documents are prepared and maintained separately:

1. Privacy Policy (NDPA/NDPR compliant, all data categories)
2. Terms of Service
3. Explicit Consent Framework (exact checkbox language per touchpoint)
4. Data Subject Rights Procedure (internal)
5. Data Protection Impact Assessment — DPIA (home visit ID verification + health data)
6. Data Retention Policy
7. Data Breach Response Plan
8. Record of Data Processing Activities (ROPA)

9. Third-Party Data Processor Checklist

10. NDPC Registration Checklist

25. TECHNICAL ARCHITECTURE

25.1 Frontend

Layer	Technology
Framework	Next.js 15 (App Router) + TypeScript
Styling	Tailwind CSS + shadcn/ui (ported from existing Lovable components)
Data fetching	TanStack Query (client/dashboard data)
Forms + validation	React Hook Form + Zod (schemas shared via <code>packages/shared</code>)
Maps	Google Maps JS API (referrer-restricted key — client-side display only)
Icons	lucide-react

25.2 Backend

Layer	Technology
Framework	NestJS (TypeScript)
Database	PostgreSQL + PostGIS (geo / "Near Me" queries)
ORM	Drizzle ORM (migration-driven schema management)
Authentication	JWT (access + refresh tokens); argon2/bcrypt password hashing; Nest guards/policies for role-based authorization (replaces Supabase RLS)
Realtime chat	Socket.io
Background jobs	BullMQ + Redis (replaces <code>pg_cron</code> — handles auto-complete-bookings, booking reminders, notification queuing)
Validation	Zod (shared schemas from <code>packages/shared</code>)
	Paystack (webhooks for booking creation, Refunds API for cancellations,

Payments	SubAccounts roadmap)
Email	Resend

25.3 Supporting Services

Service	Provider	Purpose
Public media (photos, logos)	Cloudinary	Profile photos, facility photos, logos — public CDN
Sensitive documents	Cloudflare R2 / S3	ID documents, selfies, credential certs — private bucket, signed URLs
Cache / queue	Redis	BullMQ job queue + application cache
SMS	Termii or Africa's Talking	Booking confirmations, ID verification results, time-sensitive alerts
WhatsApp	WhatsApp Business API (post-launch)	Rich booking notifications, meeting links
Error monitoring	Sentry (web + API)	Runtime error tracking and alerting
Uptime / logs	Uptime Kuma + Coolify logs	Service health and log aggregation
Product analytics	PostHog	Usage analytics, funnel analysis
Search (post-launch)	Meilisearch / Typesense	Full-text practitioner and facility search as listing volume grows
Database backups	Scheduled Postgres dumps → Cloudflare R2	Automated backup pipeline

25.4 Maps & Location Keys

Two separate Google Maps API keys:

- **Client-side key** (Maps JavaScript API): referrer-restricted to `getaidar.com` — used for embedded map display
- **Server-side key** (Directions API / Distance Matrix API / Routes API only): no referrer restriction, API-restricted to routing endpoints only — used by NestJS for ETA and route

calculations

This split prevents both the 403 "referrer <empty> blocked" routing error and exposes the client-side key only for map display.

25.5 Migration from Lovable/Supabase

The current MVP is running on Lovable.dev with a Lovable-managed Supabase instance (project ref `sgawctyyymfjtapgabwp`). The production stack above replaces this with a self-hosted, separated architecture. Migration path: export Supabase data → import to self-managed PostgreSQL with PostGIS → replace Edge Functions with NestJS controllers → replace `pg_cron` with BullMQ → replace Supabase Realtime with Socket.io → replace Supabase Auth with JWT. GitHub repository connected.

26. SECURITY

26.1 Authentication & Access Control

- Passwords hashed via `argon2/bcrypt` — no plain-text storage
- JWT access + refresh token pattern — short-lived access tokens (15 minutes), longer-lived refresh tokens
- NestJS guards enforce role-based access control at the route level — patients, practitioners, facilities, and admins have separate permission sets
- Admin routes return 403 for all non-admin roles, regardless of frontend state

26.2 Data Access Controls

- PostgreSQL row-level permissions implemented via Drizzle ORM query scoping (replaces Supabase RLS) — each user can only query their own records
- Sensitive fields (ID document URLs, selfie URLs, bank account details, credential certificate URLs) are never included in patient-facing API responses
- Private documents accessed only via short-lived signed URLs generated server-side on authenticated request

26.3 Booking Concurrency

- Database-level unique partial index prevents double-booking regardless of application-level race conditions (see Section 12.3)

26.4 Rate Limiting

- NestJS rate limiting applied to all public-facing endpoints (signup, login, search, booking creation, waitlist submission)
- Stricter limits on authentication endpoints to prevent brute-force attacks
- Redis-backed rate limiting ensures limits persist across multiple server instances

26.5 API Keys & Secrets

- All API keys stored as encrypted environment variables — never in source code
- Google Maps client key referrer-restricted, server key API-restricted
- Paystack webhook signature verification on all incoming webhook events — rejects unsigned requests

26.6 Audit Logging

- All admin actions logged to `admin_audit_log` — immutable, append-only, includes action type, target, admin ID, and payload snapshot
- Provides accountability, tamper evidence, and regulatory compliance for all data operations involving user data

26.7 Error Monitoring

- **Sentry** integrated on both Next.js frontend and NestJS backend
- Alerts on unhandled exceptions, payment failures, webhook processing errors, and high error rates
- Personally identifiable data scrubbed from Sentry payloads

26.8 Backups

- Scheduled Postgres dump jobs (BullMQ, daily minimum) → Cloudflare R2 backup bucket
- Backup integrity verified on a weekly basis
- Restore procedure tested quarterly

26.9 Uptime Monitoring

- **Uptime Kuma** monitors all critical endpoints and services
- Alerts on downtime, latency spikes, and SSL certificate expiry

- Coolify logs for server-level log aggregation

26.10 Payments Security

- Aidar does not store card details — all payment card processing is handled by Paystack
- Paystack webhook events verified via HMAC-SHA512 signature before processing
- Refunds processed only via authenticated NestJS service calls, not directly from frontend

27. BRANDING

- **Platform name:** Aidar (from “aider” — aiding healthcare access)
- **Domain:** getaidar.com (Namecheap, WhoisGuard privacy enabled)
- **Legal entity:** EML Global Ventures Limited (principal business activity covers solar, fire safety, logistics, security — amendment or new entity registration for healthcare-technology needed before investor due diligence); **Aidar Technologies Limited** incorporation in progress
- **Visual direction:** lowercase wordmark, “pin + pulse” icon, warm teal (#0F5C5C) + coral (#FF6B4A) palette, Sora headings, Manrope/Inter body
- All platform text, emails, legal pages, and metadata updated from “MedPlus” to “Aidar”

28. DATABASE SCHEMA (KEY TABLES)

Table	Purpose	Key Fields
profiles	Core user record	id, user_type, full_name, email, phone_number
practitioners	Practitioner data	specialty, qualifications, regulatory number, fees, offers_home_visits, offers_virtual, is_verified, status, bank_name, account_number, account_name, bank_details_verified, location GEOGRAPHY(Point), response_time_minutes, acceptance_rate_pct

facilities	Facility data	type , location GEOGRAPHY(Point) , services , is_verified , status , accepts_hmo , nhis
bookings	All booking records	type , booking_date , time_slot , status , payment_status , refund_status , refund_reference , patient_id , practitioner_id , booking_for (self or dependent — Section 29.2)
reviews	Patient reviews	rating , text , booking_id , status (active/removed), removed_by , removal_reason
patient_verifications	Home-visit ID records	ID type (no BVN), ID number, document_url (R2 signed), selfie_url (R2 signed), status
availability_settings	Practitioner schedule	practitioner_id , working_days[] , start_time , end_time
blocked_dates	Practitioner date blocks	practitioner_id , blocked_date , reason , range_start , range_end
conversations	Chat threads	patient_id , practitioner_id (UNIQUE pair)
messages	Chat messages	conversation_id , sender_id , sender_type , message_text , related_booking_id , read_at
payouts	Payout log	practitioner_id , amount , payout_date , reference , notes
referrals	Referral tracking (Section 29.3)	referrer_id , referred_id , referred_user_type , status , reward_type , reward_issued_at
dependents	Patient dependents (Section 29.2)	patient_id , dependent_name , relationship , date_of_birth
admin_audit_log	Immutable admin action log	admin_id , action_type , target_entity , target_id , payload_snapshot , timestamp

<code>user_roles</code>	Admin role assignments	<code>user_id, role</code>
<code>waitlist</code>	Pre-launch unified waitlist	<code>full_name, email, user_type, specialty, location, agreed_to_updates, waitlist_position</code>

All tables have row-level access controls. Sensitive document URLs are never stored in patient-accessible columns or returned in patient-scoped API responses.

29. ADDITIONAL PRODUCT FEATURES

29.1 HMO / Insurance Filter

Many Nigerians access healthcare through employer HMO schemes. The “Accepts HMO/Insurance” toggle is added to both the facility onboarding (Step 3) and search filters. HMO badge displayed on facility profile cards and on the public facility profile page.

For practitioners: same toggle added to onboarding Step 3 and visible on practitioner search cards and public profile.

This builds directly on the existing NHIS toggle infrastructure and requires minimal additional engineering.

29.2 Booking for a Dependent

Patients can book an appointment on behalf of a dependent (child, elderly parent, or other family member) without creating a separate account for them.

A “**Who is this booking for?**” toggle appears in the booking modal:

- **Myself** (default) — pre-fills patient’s own details
- **A dependent** — shows a dropdown of previously added dependents or a “Add new dependent” option collecting: dependent’s name, relationship (Child, Parent, Sibling, Spouse, Other), and date of birth

Dependents are stored in the `dependents` table linked to the parent patient account. The booking is associated with the patient account (`patient_id`) but the `booking_for` field records the dependent ID. The practitioner sees the dependent’s name and their relationship to the booking patient on the booking card.

ID verification for home visits: the account holder's ID verification covers bookings made for dependents — a separate ID is not required per dependent.

29.3 Referral Programme

A referral programme is the most cost-effective growth mechanism for a Lagos marketplace. Full mechanic details to be defined separately. Architecture stub:

```
referrals table: referrer_id, referred_id, referred_user_type (patient/practitioner),  
referral_code, status (pending/completed), reward_type, reward_value,  
reward_issued_at.
```

Each user has a unique referral code accessible from their dashboard. Sharing the referral link ([getaidar.com/join?ref=\[code\]](https://getaidar.com/join?ref=[code])) prefills the code at signup. When the referred user completes their first booking (patient) or first verified listing (practitioner), the referral is marked complete and the reward is issued.

Reward mechanic and values to be confirmed. Fraud prevention: one reward per unique phone number/device.

29.4 Trust Signals — Response Time and Acceptance Rate

Star rating alone is insufficient trust signal, especially for new practitioners with few reviews. Two additional trust metrics displayed on practitioner search cards and profiles:

Response time: Average time between booking notification and practitioner confirmation, calculated from booking records. Displayed as: "Responds in ~X hours" or "Usually responds same day."

Acceptance rate: Percentage of bookings confirmed vs cancelled by the practitioner. Displayed as: "Accepts X% of bookings."

These are calculated from the `bookings` table on a rolling 90-day window and stored as `response_time_minutes` and `acceptance_rate_pct` on the `practitioners` table, updated nightly via BullMQ job.

29.5 Admin Review Moderation

Covered in Section 19.6. Abusive, defamatory, or fraudulent reviews can be removed by admin via soft-delete with audit trail. Both parties notified. This is a legal necessity, not a nice-to-have.

30. GLOSSARY

- **MDCN** — Medical and Dental Council of Nigeria
- **HEFAMAA** — Health Facility Monitoring and Accreditation Agency (Lagos)
- **NHIS** — National Health Insurance Scheme
- **HMO** — Health Maintenance Organisation
- **NDPA** — Nigeria Data Protection Act 2023
- **NDPR** — Nigeria Data Protection Regulation 2019
- **NDPC** — Nigeria Data Protection Commission
- **DPO** — Data Protection Officer
- **ROPA** — Record of Processing Activities
- **DPIA** — Data Protection Impact Assessment
- **BVN** — Bank Verification Number (excluded from Aida ID verification — see Section 24.1)
- **PostGIS** — PostgreSQL extension for geographic object support
- **BullMQ** — Redis-backed job queue library for Node.js/NestJS
- **JWT** — JSON Web Token (authentication mechanism)
- **ETA** — Estimated Time of Arrival

Version 3 — June 2026. This document should be updated as architecture decisions are confirmed, new features are built, and compliance milestones are completed.